

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

- The first two lines of the log file show the initial outgoing request from your computer to the DNS server requesting the IP address of yummyrecipesforme.com. This request is sent in a UDP packet.
- The third and fourth lines of the log show the response to your UDP packet. In this case, the ICMP 203.0.113.2 line is the start of the error message indicating that the UDP packet was undeliverable to port 53 of the DNS server.
- The error message, "udp port 53 unreachable" is mentioned in the last line. Port 53 is a port for DNS service. The word "unreachable" in the message indicates the UDP message requesting an IP address for the domain "www.yummyrecipesforme.com" did not go through to the DNS server because no service was listening on the receiving DNS port.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

- After the source and destination IP addresses, there can be a number of additional details like the protocol, port number of the source, and flags. In the first line of the error log, the query identification number appears as: 35084. The plus sign after the query identification number indicates there are flags associated with the UDP message. The "A?" indicates a flag associated with the DNS request for an A record, where an A record maps a domain name to an IP address. The third line displays the protocol of the response message to the browser: "ICMP," which is followed by an ICMP error message.

The port noted in the error message is used for:

The port noted in the error message, **port 53**, is used for **DNS (Domain Name System)**

services. DNS translates domain names like “www.yummyrecipesforme.com” into IP addresses that computers use to communicate over the internet.

The most likely issue is:

The most likely issue is that the **DNS server is not running or is not accepting DNS requests on port 53.** Because the log repeatedly shows the error message “**udp port 53 unreachable,**” it indicates that no DNS service was listening on the destination port, causing each DNS request to fail.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

The incident occurred at **13:24:32.192571 (1:24 p.m. and 32.192571 seconds)**, according to the timestamp shown in the log file.

Explain how the IT team became aware of the incident:

The IT team became aware of the incident after reviewing network error logs that showed repeated DNS request failures and ICMP error messages related to unreachable UDP port 53.

Explain the actions taken by the IT department to investigate the incident:

The IT department analyzed the network traffic logs, reviewed the timestamps, checked the source and destination IP addresses, examined the UDP and ICMP packets, and identified the DNS-related error messages. They also observed repeated failed attempts to contact the DNS server

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

The investigation found that the computer at IP address **192.51.100.15** attempted to contact the DNS server at **203.0.113.2** using **UDP port 53**, which is used for DNS services. The DNS requests asked for an A record to resolve the domain “www.yummyrecipesforme.com” into an IP address. The DNS server responded with an ICMP error message stating “**udp port 53 unreachable,**” meaning no DNS service was listening on that port.

Note a likely cause of the incident:

A likely cause of the incident is that the DNS service on the server was down, misconfigured, blocked by a firewall, or otherwise unavailable on port 53.